Chapter 12, Problem 37P

Problem

The total power measured in a three-phase system feeding a balanced wye-connected load is 12 kW at a power factor of 0.6 leading. If the line voltage is 440 V, calculate the line current ***IL*** and the load impedance **Z*Y***.

Step-by-step solution

Step 1 of 6

Calculate the value of the line current I_L.

$$I_L = \frac{P}{\sqrt{3}V_L(\text{pf})}$$

Substitute 12 kW for P, 0.6 for pf and 440 V for V_p.

$$\begin{aligned} I_L &= \frac{12 \text{ k}}{\sqrt{3}(440)(0.6)} \\ &= \frac{12 \text{ k}}{457.26} \\ &= 26.24 \text{ A} \end{aligned}$$

Therefore, the value of the line current I_L is, $\boxed{26.24 \text{ A}}$.

Comments (1)

Anonymous

its I=P/V why multiply by the Pf ??

Step 2 of 6

Calculate the value of the power factor angle θ.

$$\theta = \cos^{-1}(\text{pf})$$

Substitute 0.6 for pf.

$$\begin{aligned} \theta &= \cos^{-1}(0.6) \\ &= 53.13° \end{aligned}$$

Therefore, the value of the power factor angle is, $53.13°$.

Step 3 of 6

Calculate the value of the apparent power S.

$$S = \frac{P}{\text{pf}}$$

Substitute 12 kW for P and 0.6 for pf.

$$\begin{aligned} S &= \frac{12\text{ k}}{0.6} \\ &= 20\text{ kW} \end{aligned}$$

Therefore, the value of the apparent power S is, 20 kW.

Step 4 of 6

Calculate the value of the reactive power Q.

$$Q = -S\sin\theta$$

Reactive power is negative since power factor is leading.

Substitute $53.13°$ for θ and 20 kW for S.

$$\begin{aligned} Q &= -(20\text{ k})(\sin 53.13°) \\ &= -(20\text{ k})(0.8) \\ &= -16\text{ kVAR} \end{aligned}$$

Therefore, the value of the reactive power is Q is, -16 kVAR.

Step 5 of 6

Consider the expression for the complex power S.

$$\mathrm{S} = P + jQ$$

Substitute -16 kVAR for Q and 12 kW for P.

$$\mathrm{S} = (12\text{ k} - j16\text{ k})\text{ VA}$$

Therefore, the value of the apparent power S is, $(12\text{ k} - j16\text{ k})\text{ VA}$.

Step 6 of 6

Consider the expression for the complex power

$$S=\frac{3(V_p)^2}{Z^*}$$

$$Z^*=\frac{3(V_p)^2}{S}$$

$$Z=\left[\frac{3(V_p)^2}{S}\right]^*$$

Substitute $(12\text{ k}-j16\text{ k})\text{ VA}$ for S and 440 V for V_p.

$$\begin{aligned}Z&=\left[\frac{3(440)^2}{12\text{ k}-j16\text{ k}}\right]^*\\&=\left[\frac{3(193600)}{12\text{ k}-j16\text{ k}}\right]^*\\&=\left[\frac{580800}{12\text{ k}-j16\text{ k}}\right]^*\\&=[17.424-j23.232]^*\end{aligned}$$

$$Z=17.424+j23.232$$

Here Z represents the total value of the load impedance. To calculate the impedance per phase of the wye load, total impedance is to be divided by 3. That is,

$$\begin{aligned}Z_{ph}&=\frac{17.424+j23.232}{3}\\&=5.808+j7.744\ \Omega\end{aligned}$$

Therefore, the value of the impedance per phase Z_{ph} is, $\boxed{5.808+j7.744\ \Omega}$.